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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA05 – COMPUTER VISION

ASSIGNMENT – 3

Feature Detection and Matching (Unit III)

Course Code	Course Name	Assessment	CO	BTL	Max. Marks
ITA05	Computer Vision	Assignment – 3	CO3	BTL4	50

CO3: Analyse and extract image features using detection and matching techniques for effective image representation and comparison.

Instructions: Answer all five questions (10 marks each). Show every computational step, state formulas before substituting values, and justify each decision using your computed results.

Mark Allocation

Q. No.	Topic	(a)	(b)	(c)	(d)	Total
1	Harris corner detection	3	3	2	2	10
2	Hough transform for lines	3	3	2	2	10
3	Principal component analysis	3	3	2	2	10
4	Scale-invariant feature matching (SIFT)	3	3	2	2	10
5	Boundary representation (chain codes)	3	3	2	2	10
	Total	15	15	10	10	50

Question 1 (CO3, BTL4, 10 Marks) – Harris Corner Analysis for Drone Visual Odometry

A delivery drone estimates its motion by tracking corners on rooftops. The Harris detector uses the structure tensor $M = \Sigma[I_x^2\ I_xI_y;\ I_xI_y\ I_y^2]$ over a 3 × 3 window with uniform weights and the response $R = \det(M) - k\cdot[\text{trace}(M)]^2$ with $k = 0.04$. A window is classified as a corner if $R > 10^6$, an edge if $R < 0$ and flat otherwise. The gradients of window W1 are given below; windows W2 and W3 have precomputed tensors $\Sigma(I_x^2, I_y^2, I_xI_y) = (4800, 4500, 300)$ and $(150, 140, 20)$.

Window W1	Row 1	Row 2	Row 3
I_x	12 15 10	14 18 11	13 16 12
I_y	11 9 14	13 17 10	12 15 13

(a) Compute ΣI_x^2, ΣI_y^2 and ΣI_xI_y for W1 and form the structure tensor M. [3 Marks]

(b) Compute det(M), trace(M), R and the eigenvalues of M for W1, and classify the window. [3 Marks]

(c) Compute R for W2 and W3, classify them, and rank W1, W2 and W3 by suitability for tracking. [2 Marks]

(d) Analyse why the Harris response is invariant to image rotation but not to scale, and the consequence for a drone that changes altitude. [2 Marks]

QUESTION 1 – Harris Corner Analysis for Drone Visual Odometry

Aim: To calculate the Harris corner response, structure tensor and eigenvalues for three image windows and identify the most suitable features for drone visual odometry.

a) Calculate the gradient sums and structure tensor for W1 (3 Marks)

Given:

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}$$

The gradient values of W1 are:

$$I_x = \begin{bmatrix} 12 & 15 & 10 \\ 14 & 18 & 11 \\ 13 & 16 & 12 \end{bmatrix}$$

$$I_y = \begin{bmatrix} 11 & 9 & 14 \\ 13 & 17 & 10 \\ 12 & 15 & 13 \end{bmatrix}$$

. Calculate \(\sum I_x^2\)

$$\begin{aligned} \sum I_x^2 &= 12^2 + 15^2 + 10^2 \\ &\quad + 14^2 + 18^2 + 11^2 \\ &\quad + 13^2 + 16^2 + 12^2 \\ &= 144 + 225 + 100 + 196 + 324 \\ &\quad + 121 + 169 + 256 + 144 \\ &= \boxed{1679} \end{aligned}$$

2. Calculate $\sum I_y^2$

$$\begin{aligned} \sum I_y^2 &= 11^2 + 9^2 + 14^2 \\ &\quad + 13^2 + 17^2 + 10^2 \\ &\quad + 12^2 + 15^2 + 13^2 \\ &= 121 + 81 + 196 + 169 + 289 \\ &\quad + 100 + 144 + 225 + 169 \\ &= \boxed{1494} \end{aligned}$$

3. Calculate $\sum I_x I_y$

$$\begin{aligned} \sum I_x I_y &= 12(11) + 15(9) + 10(14) \\ &\quad + 14(13) + 18(17) + 11(10) \\ &\quad + 13(12) + 16(15) + 12(13) \\ &= 132 + 135 + 140 + 182 + 306 \\ &\quad + 110 + 156 + 240 + 156 \\ &= \boxed{1557} \end{aligned}$$

$$\boxed{M = \begin{bmatrix} 1679 & 1557 \\ 1557 & 1494 \end{bmatrix}}$$

(b) Calculate determinant, trace, R and eigenvalues (3 Marks)

1. Determinant

$$\det(M) = M_{11}M_{22} - M_{12}^2$$

$$= 1679(1494) - 1557^2$$

$$= 2,508,426 - 2,424,249$$

$$\boxed{\det(M) = 84,177}$$

$$\text{trace}(M) = 1679 + 1494$$

$$\boxed{\text{trace}(M) = 3173}$$

3. Harris response

Given $k = 0.04$,

$$R = \det(M) - k[\text{trace}(M)]^2$$

$$R = 84,177 - 0.04(3173)^2$$

$$R = 84,177 - 402,717.16$$

$$\boxed{R = -318,540.16}$$

(c) Calculate R for W2 and W3 and rank the windows (2 Marks)

Formula:

$$R = AD - B^2 - 0.04(A + D)^2$$

Window	A	D	B
W1	1679	1494	1557
W2	4800	4500	300
W3	150	140	20

For W2:

$$\det(M) = 4800(4500) - 300^2$$

$$= 21,600,000 - 90,000 = 21,510,000$$

$$\text{trace} = 4800 + 4500 = 9300$$

[R=21,510,000-0.04(9300)^2]

[R=18,050,400]

(d) Rotation invariance and scale changes (2 Marks)

The Harris detector is rotation invariant because, under rotation, the structure tensor transforms as:

\[M'=QMQ^T \]

Here, \(Q\) is an orthogonal rotation matrix.

The determinant and trace remain unchanged under this transformation. Therefore, the Harris response remains unchanged under ideal image rotation.

However, the detector is not scale invariant because:

- The apparent size of objects changes with image scale.
- Gradient measurements change with resampling.
- A fixed-size window may capture different image structures.

Drone consequence: When the drone changes altitude, rooftops appear larger or smaller. Some corners may disappear or shift.

Solution: Use image pyramids, multi-scale feature detection and feature tracking methods to improve robustness.

Result: W1 is an edge, while W2 and W3 are corners. W2 has the highest raw Harris response and is the strongest candidate based on this measure.

Question 2 (CO3, BTL4, 10 Marks) – Hough Transform Analysis for Lane-Marking Detection

An advanced driver-assistance system detects lane markings from edge pixels in a bird's-eye-view image. A small test set of five edge points is given below. The normal-form Hough transform $\rho = x \cos\theta + y \sin\theta$ is evaluated at $\theta \in \{0°, 45°, 90°, 135°\}$, and ρ is quantised to the nearest integer when voting into the accumulator.

Point	E1	E2	E3	E4	E5
(x, y)	(2, 6)	(4, 4)	(6, 2)	(1, 7)	(3, 1)

(a) Compute ρ for every point at every θ, and present the results in a table. [3 Marks]

(b) Fill the accumulator after quantising ρ, and identify the peak. [3 Marks]

(c) Convert the peak to the line equation in image coordinates and identify the outlier. [2 Marks]

(d) Analyse the effect of the coarse θ resolution and of the ρ quantisation on this result, and explain why the Hough transform is preferred over least-squares line fitting here. [2 Marks

Aim: To calculate Hough parameters, construct an accumulator, identify the dominant lane line and analyse the effects of quantization and outliers.

(a) Calculate rho for every point at every angle (3 Marks)

Formula:

\[\rho=x\cos\theta+y\sin\theta \]

Given points:

- E1 = (2,6)
- E2 = (4,4)
- E3 = (6,2)
- E4 = (1,7)
- E5 = (3,1)

Useful trigonometric values:

$$\cos 45^\circ = \sin 45^\circ = 0.7071$$

$$\cos 135^\circ = -0.7071, \quad \sin 135^\circ = 0.7071$$

Example for E1 at 45°:

$$\rho = 2(0.7071) + 6(0.7071)$$

$$\rho = 5.6569$$

Calculated rho values:

Point	0°	45°	90°	135°
E1 (2,6)	2.000	5.657	6.000	2.828
E2 (4,4)	4.000	5.657	4.000	0.000
E3 (6,2)	6.000	5.657	2.000	−2.828
E4 (1,7)	1.000	5.657	7.000	4.243
E5 (3,1)	3.000	2.828	1.000	−1.414

b) Fill the accumulator and identify the peak (3 Marks)

The rho values are rounded to the nearest integer.

Point	0°	45°	90°	135°
E1	2	6	6	3
E2	4	6	4	0
E3	6	6	2	−3
E4	1	6	7	4
E5	3	3	1	−1

At $\theta = 45^\circ$, four points vote for $\rho = 6$.

Accumulator peak:

$$\boxed{(\theta, \rho) = (45^\circ, 6)}$$

$$\boxed{\text{Peak vote count} = 4}$$

The following table shows the peak and other votes at 45°.

Rho bin	Votes
3	1
6	4

The peak corresponds to four supporting points.

(c) Convert the peak into a line equation and identify the outlier (2 Marks)

Normal-form equation:

$$x \cos\theta + y \sin\theta = \rho$$

Substituting $\theta = 45^\circ, \rho = 6$:

$$x(0.7071) + y(0.7071) = 6$$

Multiplying by $\sqrt{2}$:

$$\boxed{x + y \approx 8.485}$$

Important quantization interpretation: The four points E1, E2, E3 and E4 lie exactly on:

$$\boxed{x + y = 8}$$

Their true rho value is:

$$\rho = \frac{8}{\sqrt{2}} = 5.657$$

This is rounded to 6.

Therefore, the actual geometric line supported by the data is:

$$\boxed{x + y = 8}$$

E5 = (3,1) does not lie on this line.

Outlier:

$$\boxed{E5 = (3, 1)}$$

(d) Analyse angle resolution, rho quantization and Hough advantages (2 Marks)

Effect of coarse theta resolution:

- Only four angles are tested.
- The actual line direction may fall between available angles.
- The estimated line orientation may be inaccurate.

Effect of rho quantization:

- Rounding merges nearby line hypotheses.
- Different lines may vote into the same bin.
- The estimated position may be slightly shifted.

Why Hough is preferred over ordinary least squares:

- Hough voting can tolerate missing edge points.

- Outliers do not necessarily dominate the peak.
- Multiple line hypotheses can be identified.
- Least-squares fitting can be affected strongly by outliers.

For stronger robustness, RANSAC can also be used.

Result: The dominant lane line is \(x+y=8\), supported by E1–E4. E5 is identified as the outlier. The peak occurs at \((45^\circ,6)\) with four votes.

Question 3 (CO3, BTL4, 10 Marks) – PCA for Dimensionality Reduction in Leaf-Disease Features

An agritech start-up classifies leaf diseases using two shape features per lesion: x_1 = lesion length (mm) and x_2 = lesion width (mm). To reduce storage and noise, the engineer analyses five training lesions with principal component analysis, using the sample covariance (divide by n − 1).

Lesion	L1	L2	L3	L4	L5
x_1	2	4	6	8	10
x_2	3	5	4	8	10

(a) Compute the mean vector and the sample covariance matrix. [3 Marks]

(b) Compute the eigenvalues and the principal eigenvector, and the percentage of variance explained by each component. [3 Marks]

(c) Project the five lesions onto PC1. [2 Marks]

(d) Analyse whether a single feature should replace the two original features, and interpret PC1 physically. [2 Marks]

Aim: To calculate the mean, covariance matrix, eigenvalues, principal component and projected scores for leaf-lesion features using PCA.

(a) Calculate the mean vector and sample covariance matrix (3 Marks)

Given data:

Lesion	Length x_1	Width x_2
L1	2	3
L2	4	5
L3	6	4
L4	8	8
L5	10	10

Mean length:

$$\bar{x}_1 = \frac{2+4+6+8+10}{5}$$

$$\boxed{\bar{x}_1 = 6}$$

Mean width:

$$\bar{x}_2 = \frac{3+5+4+8+10}{5}$$

$$\boxed{\bar{x}_2 = 6}$$

Mean vector:

$$\boxed{\mu = \begin{bmatrix} 6 \\ 6 \end{bmatrix}}$$

3. Sample covariance matrix

Formula:

$$C = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \mu)(x_i - \mu)^T$$

Here $n = 5$, so the divisor is 4.

Variance of length:

$$\mathrm{Var}(x_1) = \frac{16 + 4 + 0 + 4 + 16}{4}$$

$$\boxed{\mathrm{Var}(x_1) = 10}$$

Variance of width:

$$\mathbf{Var}(x_2) = \frac{9 + 1 + 4 + 4 + 16}{4}$$

$$\boxed{\mathbf{Var}(x_2) = 8.5}$$

Covariance:

$$\mathbf{Cov}(x_1, x_2) = \frac{12 + 2 + 0 + 4 + 16}{4}$$

$$\boxed{\mathbf{Cov}(x_1, x_2) = 8.5}$$

Therefore,

$$\boxed{C = \begin{bmatrix} 10 & 8.5 \\ 8.5 & 8.5 \end{bmatrix}}$$

(b) Calculate eigenvalues and principal eigenvector (3 Marks)

The characteristic equation is:

$$|C - \lambda I| = 0$$

$$\begin{vmatrix} 10 - \lambda & 8.5 \\ 8.5 & 8.5 - \lambda \end{vmatrix} = 0$$

$$(10 - \lambda)(8.5 - \lambda) - 72.25 = 0$$

$$\lambda^2 - 18.5\lambda + 12.75 = 0$$

Solving:

$$\boxed{\lambda_1 = 17.7830}$$

$$\boxed{\lambda_2 = 0.7170}$$

Percentage of variance explained

Total variance:

$$17.7830 + 0.7170 = 18.5$$

PC1:

$$\frac{17.7830}{18.5} \times 100$$

$$\boxed{PC1 = 96.12\%}$$

PC2:

$$\frac{0.7170}{18.5} \times 100$$

$$\boxed{PC2 = 3.88\%}$$

(c) Project the five lesions onto PC1 (2 Marks)

Projection formula:

$$z_i = (x_i - \mu)^T v_1$$

Here:

$$v_1 = \begin{bmatrix} 0.7375 \\ 0.6753 \end{bmatrix}$$

For L1:

$$z_1 = (-4)(0.7375) + (-3)(0.6753)$$

$$\boxed{z_1 \approx -4.9761}$$

(d) Should one feature replace the original two features? (2 Marks)

PC1 explains approximately 96.12% of the total sample variance.

This indicates that most variation in the two original measurements can be represented using a single component.

Physical interpretation of PC1:

PC1 represents a combined lesion size direction because lesion length and width increase together.

Advantages:

- Reduces storage requirements.
- Reduces feature dimensionality.
- May reduce noise and redundancy.
- Simplifies downstream analysis.

Limitations:

- PCA is unsupervised.
- High variance does not guarantee disease-class separation.
- Some disease-specific information may be lost.

Result: PC1 can be used as a compact representation because it explains 96.12% of variance.

However, classification performance should be validated before replacing the two original features.

Question 4 (CO3, BTL4, 10 Marks) – SIFT Scale-Space and Descriptor Matching for Product Recognition

A retail app recognises packaged products on shelves by matching SIFT features against a product database. The detector uses a base scale $\sigma_0 = 1.6$ with S = 3 intervals per octave. For fast analysis, the 128-dimensional descriptors are represented here by 4-dimensional summaries. Three query descriptors from a shelf photo and three database descriptors are given. Lowe's ratio test accepts a match if $d_1/d_2 < 0.8$, where d_1 and d_2 are the nearest and second-nearest Euclidean distances.

Descriptor	Values
D1 (database)	14, 38, 9, 22
D2 (database)	30, 12, 20, 19
D3 (database)	11, 44, 4, 30
Q1 (query)	12, 40, 7, 25
Q2 (query)	30, 10, 22, 18
Q3 (query)	12.5, 41, 6.5, 26

(a) Compute $k = 2^{1/S}$, the scales of the S + 3 Gaussian images in the first octave, the number of DoG images, and the length of a full SIFT descriptor. [3 Marks]

(b) Compute the Euclidean distances from Q1, Q2 and Q3 to D1, D2 and D3. [3 Marks]

(c) Apply the ratio test to each query and state the accepted matches. [2 Marks]

(d) Analyse why the ratio test is more reliable than a fixed distance threshold, using Q3 as evidence, and why SIFT tolerates shelf photos taken at different distances. [2 Marks]

(a) Calculate scale factor, Gaussian images, DoG images and descriptor length (3 Marks)

Given:

- Base scale \(\sigma_0=1.6\)
- Intervals per octave \(S=3\)

1. Scale factor

$$k = 2^{1/S}$$

$$k = 2^{1/3}$$

$$\boxed{k = 1.2599}$$

2. Gaussian scale calculation

$$\sigma_i = \sigma_0 k^i$$

$$\sigma_i = 1.6(2^{i/3})$$

$$\overline{\text{Gaussian images} = S + 3 = 6}$$

3. Number of DoG images

Difference of Gaussian:

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

The number of adjacent differences from six Gaussian images is:

$$\boxed{6 - 1 = 5 \text{ DoG images}}$$

(b) Calculate Euclidean distances (3 Marks)

Formula:

$$d(A,B) = \sqrt{\sum_{j=1}^{n}(A_j - B_j)^2}$$

Database descriptors:

$$D_1 = (14, 38, 9, 22)$$

$$D_2 = (30, 12, 20, 19)$$

$$D_3 = (11, 44, 4, 30)$$

Query descriptors:

$$Q_1 = (12, 40, 7, 25)$$

$$Q_2 = (30, 10, 22, 18)$$

$$Q_3 = (12.5, 41, 6.5, 26)$$

Q1 distances

$$d(Q_1, D_1) = \sqrt{(-2)^2 + 2^2 + (-2)^2 + 3^2}$$

$$= \sqrt{21} = \boxed{4.5826}$$

$$d(Q_1, D_2) = \boxed{36.2353}$$

$$d(Q_1, D_3) = \boxed{7.1414}$$

Q2 distances

$$d(Q_2, D_1) = \boxed{35.0000}$$

Q3 distances

$$d(Q_3, D_1) = \boxed{5.7879}$$

$$d(Q_3, D_2) = \boxed{37.1282}$$

$$d(Q_3, D_3) = \boxed{5.7879}$$

(c) Apply Lowe's ratio test (2 Marks)

Formula:

$$\text{Ratio} = \frac{d_1}{d_2}$$

Accept if:

$$\frac{d_1}{d_2} < 0.8$$

Query Q1

Nearest match: D1

$$d_1 = 4.5826, \quad d_2 = 7.1414$$

$$\text{Ratio} = \frac{4.5826}{7.1414}$$

$$\boxed{\text{Ratio} = 0.6415}$$

Since 0.6415 is less than 0.8:

Q1 → D1 accepted.

Query Q2

Nearest match: D2

$$d_1 = 4.2426, \quad d_2 = 35$$

$$\boxed{\text{Ratio} = 0.1212}$$

Since 0.1212 is less than 0.8:

Q2 → D2 accepted.

Query Q3

Nearest distance is tied:

$$ d(Q_3,D_1)=d(Q_3,D_3)=5.7879 $$

Thus:

$$ d_1=d_2=5.7879 $$

$$ \boxed{\text{Ratio}=1} $$

(d) Reliability of ratio test and scale tolerance (2 Marks)

The ratio test measures the distinctiveness of the nearest match compared with the second-nearest match.

Why it is more reliable than a fixed threshold:

- A fixed threshold considers only absolute distance.
- The ratio test checks whether the best candidate is clearly better than alternatives.
- Ambiguous matches can be rejected.

Q3 interpretation:

Q3 has equal distances to D1 and D3.

Even though these distances are relatively small, both candidates are equally close. Therefore, the ratio is 1, and the match is rejected.

This prevents uncertain product recognition.

Why SIFT tolerates different distances:

- SIFT detects features across Gaussian scale-space.
- It identifies keypoints at different scales.
- Orientation assignment helps rotation tolerance.
- Normalized local gradient descriptors improve robustness to certain illumination changes.

However, SIFT is not perfectly invariant to extreme perspective, occlusion or severe blur.

Result: Q1 matches D1 and Q2 matches D2. Q3 is rejected because two database descriptors are equally close, making the match ambiguous.

Question 5 (CO3, BTL4, 10 Marks) – Chain-Code Representation for Machined-Part Shape Inspection

A machine shop verifies the profile of punched sheet-metal parts using boundary descriptors that must be independent of the part's position and orientation on the conveyor. The traced boundary of a part, with x to the right and y upward, visits the grid points (0, 0) → (1, 0) → (2, 0) → (3, 1) → (3, 2) → (2, 3) → (1, 3) → (0, 2) → (0, 1) → (0, 0). The 8-direction Freeman code is used: 0 = east, 1 = north-east, 2 = north, 3 = north-west, 4 = west, 5 = south-west, 6 = south and 7 = south-east.

(a) Derive the 8-direction Freeman chain code of the boundary and normalise it with respect to the starting point. [3 Marks]

(b) Compute the circular first difference and the shape number. [3 Marks]

(c) Compute the perimeter from the chain code and the enclosed area of the polygon. [2 Marks]

(d) The part is rotated by 90° counter-clockwise on the conveyor. Derive the new chain code and first difference, and analyse which descriptor should be stored as the inspection template. [2 Marks]

1. Perimeter

- Orthogonal steps = 6
- Diagonal steps = 3

Formula:

$$P = N_o + N_d\sqrt{2}$$

$$P = 6 + 3\sqrt{2} = \boxed{10.24 \text{ units}}$$

2. Area

Using the boundary coordinates and the Shoelace formula:

$$A = \frac{1}{2}\left|\sum x_i y_{i+1} - y_i x_{i+1}\right|$$

Area = 7.5 square units.

3. First Difference Chain Code

Formula:

$$d_i = (c_{i+1} - c_i) \bmod 8$$

First difference: `0, 1, 1, 1, 1, 1, 1, 0, 2`

4. Shape Recognition

- **Rotation invariance:** Use direction differences modulo 8.
- **Starting-point invariance:** Use the minimum circular shift (canonical form).
- Chain code represents the object's boundary and helps detect shape defects.

Result: The chain code helps inspect machined parts by representing boundary shape, perimeter and shape features.